CivicPulse - Data Dictionary

The database `nyc311` holds 45,750 NYC 311 service requests from 2024 in 4 tables: one main table of requests, and three small tables that describe who handles them, what the problem is, and where it happened. Source: NYC Open Data, *311 Service Requests from 2020 to Present* (`erm2-nwe9`).

1. The requests: `service_request`

One row per 311 request, 45,750 rows; it holds the dates used for every time and response-time KPI, and points to the three other tables.

Column	Type	Key	Null allowed	Meaning	Example	Source column
request_id	BIGINT	PK	No	311 request number	60123456	unique_key
created_date	TIMESTAMP		No	When the resident reported the problem	2024-03-05 14:22:10	created_date
closed_date	TIMESTAMP		Yes	When the agency closed the request; empty if still open	2024-03-06 09:10:00	closed_date
due_date	TIMESTAMP		Yes	Agency deadline; 99.5% empty, not used	—	due_date
status	VARCHAR(30)		Yes	Current state of the request	Closed	status
channel	VARCHAR(20)		Yes	How the request was submitted	MOBILE	open_data_channel_type
resolution_description	TEXT		Yes	What the agency did	The Police Department responded…	resolution_description
latitude	NUMERIC(9,6)		Yes	GPS latitude of the incident	40.693900	latitude
longitude	NUMERIC(9,6)		Yes	GPS longitude of the incident	-73.989700	longitude

Column	Type	Key	Null allowed	Meaning	Example	Source column
agency_code	VARCHAR(20)	FK → agency	No	Agency in charge	NYPD	agency
complaint_type_id	INT	FK → complaint_type	No	Type of problem	7	complaint_type + descriptor
location_id	INT	FK → location	No	Place of the incident	42	incident_zip + borough

Indexes: created_date, agency_code, complaint_type_id, location_id (plus the primary key).

2. The 3 other tables

Each of these tables stores a name only once, and service_request points to it by its key, so names are not repeated on thousands of rows.

agency — who handles the request (14 rows)

Column	Type	Key	Null allowed	Meaning	Example
agency_code	VARCHAR(20)	PK	No	Short agency code	NYPD
agency_name	TEXT		No	Full agency name	New York City Police Department

complaint_type — what the problem is (709 rows)

Column	Type	Key	Null allowed	Meaning	Example
complaint_type_id	SERIAL	PK	No	Number generated by PostgreSQL	7
complaint_type	TEXT		No	Main category	Noise - Residential
descriptor	TEXT		No	Detail of the problem; Unspecified if missing in the source	Loud Music/Party

The pair (complaint_type, descriptor) is unique.

location — where it happened (222 rows)

Column	Type	Key	Null allowed	Meaning	Example
location_id	SERIAL	PK	No	Number generated by PostgreSQL	42
incident_zip	VARCHAR(10)		No	ZIP code; Unknown if missing in the source	11201
borough	VARCHAR(20)		No	BROOKLYN, QUEENS, MANHATTAN, BRONX, STATEN ISLAND, or Unspecified	BROOKLYN

The pair (incident_zip, borough) is unique.

3. Relationships, views and data quality

All three relationships are one-to-many: one agency, complaint type or location appears in many requests.

Relationship	From	To	Cardinality
Handled by	service_request.agency_code	agency.agency_code	many → 1
Describes	service_request.complaint_type_id	complaint_type.complaint_type_id	many → 1
Located at	service_request.location_id	location.location_id	many → 1

Relational notation:

- agency (**agency_code**, agency_name)
- complaint_type (**complaint_type_id**, complaint_type, descriptor)
- location (**location_id**, incident_zip, borough)
- service_request (**request_id**, created_date, closed_date, due_date, status, channel, resolution_description, latitude, longitude, #agency_code, #complaint_type_id, #location_id)

(**bold** = primary key, # = foreign key)

Views for Power BI (views.sql):

View	Content	Main columns
v_requests	One row per request, with the agency, problem and place names	month, hour_of_day, weekday_number, resolution_time
v_kpi_overview	One row: the headline KPIs	total_requests, open_requests, closure_rate_pct, median_resolution_time

View	Content	Main columns
v_data_quality	Number of requests affected by each data problem	check_name, rows_affected
v_requests_by_hour	Requests per hour of the day	requests, pct_of_requests
v_requests_by_weekday	Requests per day of the week	requests, pct_of_requests
v_requests_by_weekday_hour	Requests per day and hour (heatmap)	requests
v_busiest_hour_by_type	Busiest hour of the 5 main problems	busiest_hour, requests
v_monthly_requests	Requests per month and change from the previous month	total_requests, net_change
v_requests_by_borough	Requests per borough and share of the city	total_requests, request_pct
v_open_requests_by_borough	Open requests per borough	open_requests, open_rate_pct
v_top_complaint_types	10 most common problems	request_count
v_top3_types_by_borough	3 most common problems in each borough	rank, requests
v_top_zip_codes	10 ZIP codes with the most requests	request_count
v_resolution_by_type	Typical solving time per problem	nb_closed, median_hours, median_days
v_resolution_by_agency	Typical solving time and volume per agency	nb_requests, nb_closed_valid, median_hours, median_days
v_open_closed_by_agency	Open and closed requests per agency	open_requests, closed_requests, pct_open
v_solved_within_by_agency	Share solved within 24 hours and within 7 days per agency	nb_closed_valid, pct_24h, pct_7d

resolution_time = closed_date − created_date, only when closed_date is not empty and not before created_date (empty otherwise). Every solving-time column is built on it: nb_closed_valid counts the requests where it exists, and median_hours / median_days are its median.

Data-quality notes:

Column	Issue	Share	Handling
due_date	Empty	99.5%	Not used; no "overdue" KPI
closed_date	Empty	1.7%	Excluded from resolution time

Column	Issue	Share	Handling
status, closed_date	Status Closed but no closing date	331 rows, all from DHS (331 of its 685 requests)	Excluded from resolution time; DHS solving time and % within 24 h / 7 days cover only its 354 other requests, so they are partial
closed_date	Before created_date	10 rows: 9 DOT requests with status Pending, 1 NYPD request with status Closed	Excluded from resolution time
latitude, longitude	Empty	1.5%	Left out of maps
descriptor	Empty in source	3.2%	Stored as Unspecified
incident_zip	Empty in source	0.9%	Stored as Unknown
borough	Unspecified	28 rows	Left out of borough charts

Shares come from the sample check; data_quality_checks.sql gives the exact counts in the database.